

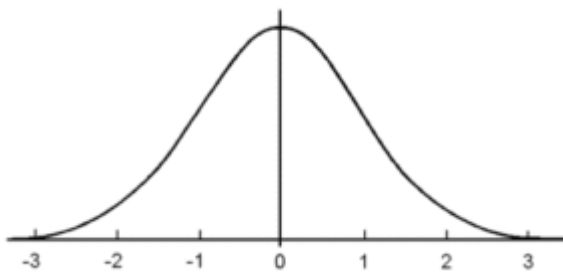
## Q)The Normal Distribution and the Empirical Rule

### Learning Objectives

- What is a normal distribution?
- The Empirical Rule (the 68 – 95 – 99.7 Rule)
- Drawing a normal distribution for a particular scenario
- Finding the percentage as it pertains to a particular scenario

### What is a normal distribution?

Early statisticians noticed the same shape coming up over and over again in different distributions—so they named it the normal distribution.



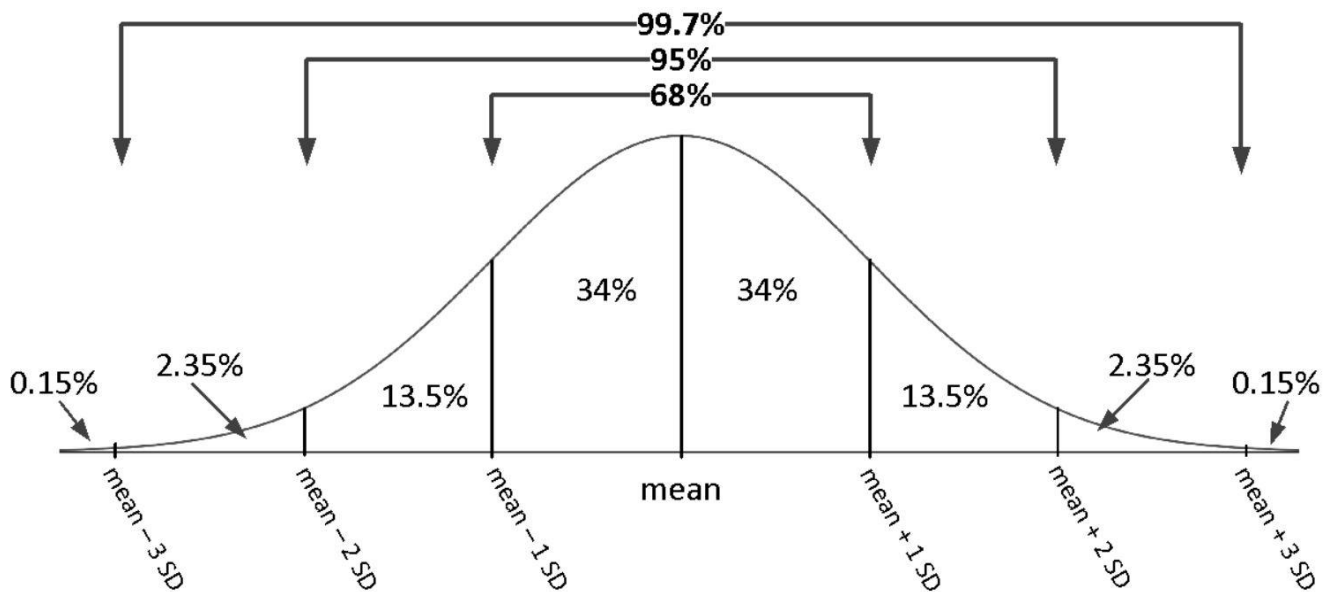
### Features of the Normal Distribution

- It is bell-shaped
- It is symmetric about the mean. In other words, if you divide the graph in half by drawing a vertical line at the mean, there will be two identical halves
- The mean, median and mode are equal and located at the center of the distribution
- It has no gaps or holes so we would say the distribution is continuous and extends out along the horizontal axis from  $-\infty$  to  $+\infty$
- An area under the distribution that falls between two data values is the same as the percentage of the data (in decimal form) between the same data values
- The total area under a normal distribution is exactly 1 or 100%

### THE EMPIRICAL RULE

For data that is bell-shaped and symmetrical (Normal):

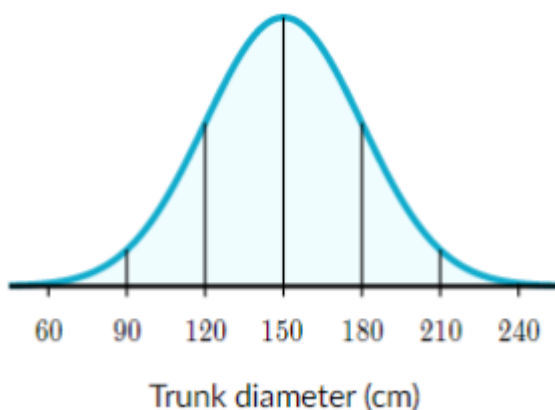
- Approximately 68% of the data is within  $\pm 1$  standard deviations of the mean.
- Approximately 95% of the data is within  $\pm 2$  standard deviations of the mean.
- Approximately 99.7% of the data is within  $\pm 3$  standard deviations of the mean.



### Example – Drawing a Normal Distribution

The trunk diameter of a certain variety of pine tree is normally distributed with a mean of  $\mu=150\text{cm}$  and a standard deviation of  $\sigma=30\text{cm}$ . Sketch a normal curve that describes this distribution.

- Step 1: Sketch a blank normal distribution
- Step 2: The mean of 150 cm goes in the middle
- Step 3: Label each standard deviation. Each standard deviation is a distance of 30 cm



### Example – Finding Percentages

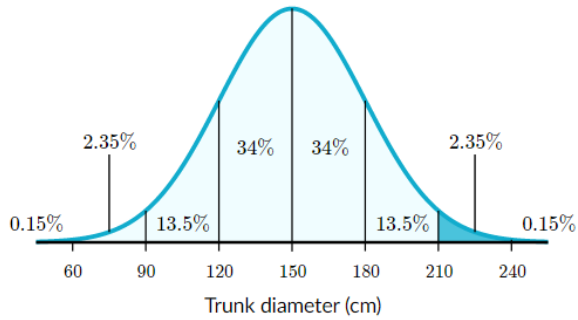
A certain variety of pine tree has a mean trunk diameter of  $\mu=150\text{cm}$  and a standard deviation of  $\sigma=30\text{cm}$

- Approximately what percent of these trees have a diameter greater than 210?
- Approximately what percent of these trees have a diameter between 90 and 210 centimeters?

a)

- Step 1: Sketch a normal curve with a mean of 150 cm and a standard deviation of 30 cm

- Step 2: Shade the region corresponding to a diameter greater than 210 cm.
- Step 3: Add the percentages in the shaded area



$$2.35\% + 0.15\% = 2.5\%$$

About 2.5% of these trees have a diameter greater than 210 cm.

b)

- Step 1: Use the same distribution from part a
- Step 2: Shade the region between 90 and 210 cm
- Step 3: Add the percentages in the shaded area.
- $13.5\% + 34\% + 34\% + 13.5\% = 95\%$
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- **About 95% of these trees have a diameter between 90 and 210 cm.**